Bohan Wang

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OBJECTIVE: Full-time in digital ASIC/FPGA engineering, with interest in verification, DFT and CAD **(visa sponsorship required)**

EDUCATION

University of Michigan, Ann Arbor, MI

Master of Science in Electrical and Computer Engineering, 3.4/4.0 Integrated Systems & VLSI Expected Dec 2017

- Related Coursework: Compiler, Advanced Computer Architecture, HW & SW Formal Verification, Digital Testing, VLSI Design, Computer Architecture, Solid-State Device Lab, Control System Analysis & Design, Parallel Architecture

University of Washington, Seattle WA

Bachelor of Science in Electrical Engineering, 3.52/4.0, Six quarters of Dean's List Sept 2011 to June 2015

- Related Coursework: Digital VLSI design, Analog Circuit Design, Hardware/Software Interface, Data Structure and Algorithm, Embedded Microcomputer Systems, Semiconductor Devices, Digital Signal Processing, Applied Electromagnetics

PROJECT EXPERIENCE

UVM Verification Environment of Asynchronous FIFO (Personal Project) May 2017 to Aug 2017

- Study and develop a basic verification environment from scratch using **UVM** on asynchronous FIFO and simulate with ModelSim
- Utilize one environment, agent, driver, sequencer, two monitors and multiple sequences for different verification scenarios
- Use SVA on reset and **constraint-random tests** to randomize clock frequencies and number of write/read ops per iteration

COOL Compiler Implementation June 2017 to Aug 2017

- Implement a compiler for a strongly-typed object-oriented language in C++ to generate MIPS assembly simulated on SPIM
- Implement compiler front end along with Flex/Bison with multiple-pass on semantic checking and type recovery

Formal Verification of RTL in Symbolic Trajectory Evaluation (STE) Feb 2017 to Apr 2017

- Researched, compared and surveyed various literature on STE efficiency and effectiveness in NuSMV and Forte environment
- Implement formal verification on hardware benchmarks such as ALU and memory systems

Configurable and Energy-efficient JPEG compression ASIC with Approximate Computing (VLSI Design) Jan 2017 to Apr2017

- Designed **hierarchical** JPEG compression ASIC core with datapath pipeline in architectural and cell-level optimizations
- Achieved configurable energy-quality tradeoff using **custom approximate adder cells** and **voltage scaling** through digital LDO
- Underwent physical design flow in Innovus with 64 IO pads and 12.6mm^2, with maximum **30% energy reduction** at 200MHz

Two-way Superscalar Out-of-Order R10k RISC Processor Design (Computer Architecture) Sept 2016 to Dec 2016

- Designed synthesizable **two-way out-of-order** Alpha processor in R10K microarchitecture in Verilog running at 100MHz
- Improved performances with **fast branch recovery up** to four branches, **load-store queue** and **two-way associative data cache**
- Verified and tested extensively using interactive visual debugger and self-checking testbench, and **passed all benchmarks**

Low Power ATPG Implementation in Genetic Algorithm (DFT Project) Nov 2016 to Dec 2016

- Designed and implemented a **low power ATPG improvement** in C++ in genetic algorithm on a PODEM implementation
- Implemented **greedy algorithm** to reduce switching activity by **40% in average** for low power test pattern generation

WORK EXPERIENCE

Lab/Research Assistant, Prof. Zhengya Zhang's VLSISP Lab

University of Michigan, Department of Electrical Engineering and Computer Science Jun 2017 to Present

- Assist in automation flow in Python of dynamic energy extraction of various RTL template on TSMC 40nm and TSMC 16nm
- Assist in developing and testing Verilog and Python code of scan chain on Opal Kelly prototype board with Xilinx FPGA

Analog/RF CAD Engineering Intern

Qualcomm Inc., **(Former Atheros),** San Jose, CA May 2016 to Aug 2016

- Integrated and augmented constraint-driven tools in Virtuoso with custom SKILL/Perl code for analog layout rapid prototyping
- Incorporated Modgen constraints and PCell library as the foundations of layout automation flow
- Worked with circuit designers and CAD engineers to get feedback and improve user experience on CAD flow

Firmware Testing Intern

Hangzhou Trustnetic Co., LTD., Hangzhou, Zhejiang, China July 2015 to Aug 2015

- Developed testing program of firmware for data packets communication on a Motorola M-CORE processor
- Evaluated firmware functionality assisted by communication protocol with testbench via shared mailboxes by semaphores

SKILLS

- Programming languages: Java, SystemVerilog, C/C++, Python, Perl, MATLAB, Lisp, JavaScript
- EDA Tools: Synopsys (HSPICE, VCS, Design Compiler, IC Compiler), Cadence (Virtuoso Suite, Encounter, Incisive, Innovus), Mentor Graphics (nmLVS, nmDRC)